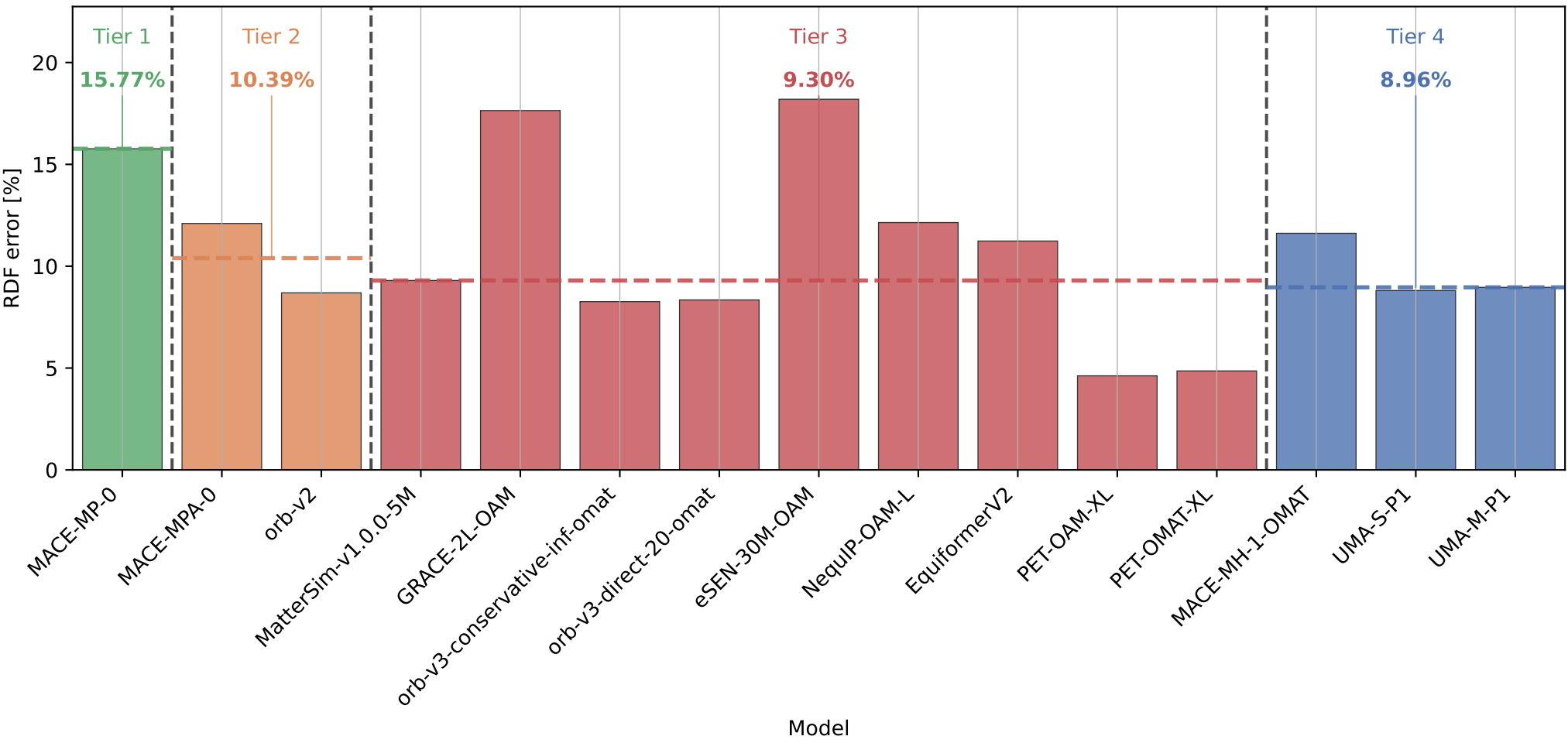
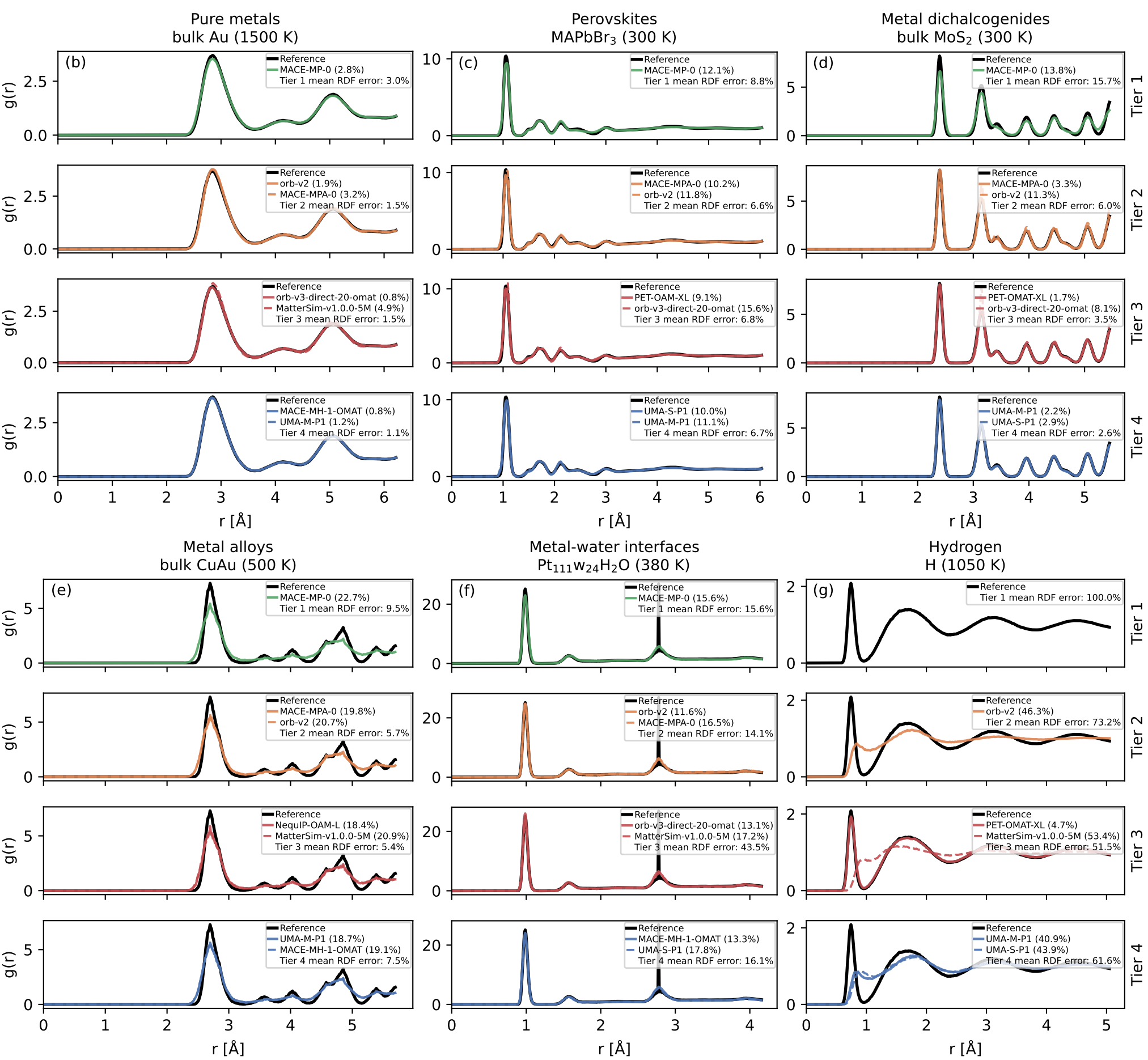
(a)


Tier 1
15.77%
Tier 2
10.39%
Tier 3
9.30%
Tier 4
8.96%
RDF error [%]
0
5
10
15
20
MACE-MP-0
MACE-MPA-0
orb-v2
MatterSim-v1.0.0-5M
GRACE-2L-OAM
orb-v3-conservative-inf-omat
orb-v3-direct-20-omat
eSEN-30M-OAM
NequIP-OAM-L
EquiformerV2
PET-OAM-XL
PET-OMAT-XL
MACE-MH-1-OMAT
UMA-S-P1
UMA-M-P1
Model


Pure metals
bulk Au (1500 K)
Perovskites
MAPbBr3 (300 K)
Metal dichalcogenides
bulk MoS2 (300 K)
(b)
Reference
MACE-MP-0 (2.8%)
Tier 1 mean RDF error: 3.0%
(c)
Reference
MACE-MP-0 (12.1%)
Tier 1 mean RDF error: 8.8%
(d)
Reference
MACE-MP-0 (13.8%)
Tier 1 mean RDF error: 15.7%
Tier 1
Reference
orb-v2 (1.9%)
MACE-MPA-0 (3.2%)
Tier 2 mean RDF error: 1.5%
Reference
MACE-MPA-0 (10.2%)
orb-v2 (11.8%)
Tier 2 mean RDF error: 6.6%
Reference
MACE-MPA-0 (3.3%)
orb-v2 (11.3%)
Tier 2 mean RDF error: 6.0%
Tier 2
Reference
orb-v3-direct-20-omat (0.8%)
MatterSim-v1.0.0-5M (4.9%)
Tier 3 mean RDF error: 1.5%
Reference
PET-OAM-XL (9.1%)
orb-v3-direct-20-omat (15.6%)
Tier 3 mean RDF error: 6.8%
Reference
PET-OMAT-XL (1.7%)
orb-v3-direct-20-omat (8.1%)
Tier 3 mean RDF error: 3.5%
Tier 3
Reference
MACE-MH-1-OMAT (0.8%)
UMA-M-P1 (1.2%)
Tier 4 mean RDF error: 1.1%
Reference
UMA-S-P1 (10.0%)
UMA-M-P1 (11.1%)
Tier 4 mean RDF error: 6.7%
Reference
UMA-M-P1 (2.2%)
UMA-S-P1 (2.9%)
Tier 4 mean RDF error: 2.6%
Tier 4
g(r)
r [Å]
Metal alloys
bulk CuAu (500 K)
Metal-water interfaces
Pt111w24H2O (380 K)
Hydrogen
H (1050 K)
(e)
Reference
MACE-MP-0 (22.7%)
Tier 1 mean RDF error: 9.5%
(f)
Reference
MACE-MP-0 (15.6%)
Tier 1 mean RDF error: 15.6%
(g)
Reference
Tier 1 mean RDF error: 100.0%
Tier 1
Reference
MACE-MPA-0 (19.8%)
orb-v2 (20.7%)
Tier 2 mean RDF error: 5.7%
Reference
orb-v2 (11.6%)
MACE-MPA-0 (16.5%)
Tier 2 mean RDF error: 14.1%
Reference
orb-v2 (46.3%)
Tier 2 mean RDF error: 73.2%
Tier 2
Reference
NequIP-OAM-L (18.4%)
MatterSim-v1.0.0-5M (20.9%)
Tier 3 mean RDF error: 5.4%
Reference
orb-v3-direct-20-omat (13.1%)
MatterSim-v1.0.0-5M (17.2%)
Tier 3 mean RDF error: 43.5%
Reference
PET-OMAT-XL (4.7%)
MatterSim-v1.0.0-5M (53.4%)
Tier 3 mean RDF error: 51.5%
Tier 3
Reference
UMA-M-P1 (18.7%)
MACE-MH-1-OMAT (19.1%)
Tier 4 mean RDF error: 7.5%
Reference
MACE-MH-1-OMAT (13.3%)
UMA-S-P1 (17.8%)
Tier 4 mean RDF error: 16.1%
Reference
UMA-M-P1 (40.9%)
UMA-S-P1 (43.9%)
Tier 4 mean RDF error: 61.6%
Tier 4
g(r)
r [Å]